

1. Counting possibilities

def. Let Ω be the set of possible outcomes of a random experiment. We call Ω the sample space.

- How do we count the number of possible outcomes?

Basic law of counting:

An experiment has n outcomes.

Another independent experiment has m outcomes.

$\Rightarrow$ the two experiments together have $n \cdot m$ outcomes.

Formally,

Ω_1 = sample space of first experiment, $|\Omega_1| = n$

Ω_2 = sample space of second experiment, $|\Omega_2| = m$

$$\Omega = \{(\omega_1, \omega_2) : \omega_1 \in \Omega_1, \omega_2 \in \Omega_2\} \Rightarrow |\Omega| = |\Omega_1| \cdot |\Omega_2| = n \cdot m$$

For further calculations we need some combinatorics.

1.1 Combinatorics

1/ Permutation: $\Omega = \{\text{all possible orderings of the elements of a set } H\}$

o Without repetition:

$H = \{h_1, \dots, h_n\}$ is a set of distinct elements.

Possible orderings: $\begin{array}{ccccccc} \square & \square & \square & \dots & \square & \square & \\ \uparrow & \uparrow & & & \uparrow & \uparrow & \\ n & (n-1) & & & 2 & 1 & \end{array} \Rightarrow |\Omega| = n(n-1) \dots 2 \cdot 1 = n!$
("n factorial", $0! = 1$)

Example: In a competition there are 4 horses.

How many possible ends does the race have? $4 \cdot 3 \cdot 2 \cdot 1 = \underline{4!} = 24$

o With repetition:

$H = \{\underbrace{h_1, \dots, h_1}_{n_1}, \underbrace{h_2, \dots, h_2}_{n_2}, \dots, \underbrace{h_r, \dots, h_r}_{n_r}\}$ is a multiset.

For $i \neq j$ the elements $h_i \neq h_j$ are different, but their multiplicity n_i might be bigger than 1. Let $n_1 + n_2 + \dots + n_r = n$.

$|\Omega| = \frac{n!}{n_1! \dots n_r!}$ $\leftarrow$ all possible ordering of n elements.
 $\leftarrow$ permuting identical elements does not give a new outcome.

Example: How many different arrangements are of the letters PEPPER?

6 letters in total

multiplicities: $P = 3 \Rightarrow \frac{6!}{3! 2! 1!} = 60$
 $E = 2$
 $R = 1$

2/ Variation: We have a set (or multiset) H of n elements, and $k \leq n$.

$\Omega = \{\text{all possible } k \text{ length words made of the elements of } H\}$

o Without repetition:

$H = \{h_1, \dots, h_n\}$ is a set of n distinguishable elements.

Possible orderings: $\begin{array}{ccccccc} \square & \square & \dots & \square & \leftarrow & \text{we only have } k \text{ positions now} \\ \uparrow & \uparrow & & \uparrow & & \\ n & (n-1) & \dots & (n-k+1) & \Rightarrow |\Omega| = n(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!} \end{array}$

Example: In a competition there are 7 horses. How many possible outcomes are there on the podium at the end of the race?

$$\begin{array}{ccc} \square & \square & \square \\ \uparrow & \uparrow & \uparrow \\ 7 & 6 & 5 \end{array} \Rightarrow 7 \cdot 6 \cdot 5 = 210 = \frac{7!}{4!}$$

o With repetition: $H = \{h_1, \dots, h_n\}$ is the same, but now each element might appear multiple times in a word.

Possible orderings: $\begin{array}{ccc} \square & \square & \dots & \square \\ \uparrow & \uparrow & & \uparrow \\ n & n & \dots & n \end{array} \leftarrow \begin{array}{l} \text{still just } k \leq n \text{ positions} \\ \text{we might repeat the same element} \end{array}$

Example:

$$\Rightarrow |\Omega| = \underbrace{n \cdot \dots \cdot n}_k = n^k$$

How many different hungarian license plates are there? letters of the english alphabet

$$\begin{array}{ccc} \square & \square & \square \\ \uparrow & \uparrow & \uparrow \\ 26 & 26 & 26 \end{array} - \begin{array}{ccc} \square & \square & \square \\ \uparrow & \uparrow & \uparrow \\ 10 & 10 & 10 \end{array} \Rightarrow 26^3 \cdot 10^3 - 26^3$$

ooo is not allowed

3/ Combination: Again, a set (or multiset) H of n elements, and $k \leq n$ are given. $\Omega = \{\text{all possible } k \text{ element groups that can be formed using the elements of } H\}$

$\Omega = \{E: E \subseteq H, |E| = k\}$, so the order of the elements chosen doesn't matter.

o Without repetition:

$H = \{h_1, \dots, h_n\}$ is a set of distinguishable elements.

Possible ordering: There are $\frac{n!}{(n-k)!}$ ways of choosing an arrangement of length k .
 $\Rightarrow$ We do not count the different ordering of elements within the chosen ones, so

$$|\Omega| = \frac{n!}{(n-k)!k!} = \binom{n}{k} \quad ("n \text{ choose } k")$$

Note: $\frac{n!}{(n-k)!k!} = \binom{n}{k}$ is called binomial coefficient, by definition $\binom{n}{n} = \binom{n}{0} = 1$, $k > n \Rightarrow \binom{n}{k} = 0$.

For $n_1 + \dots + n_r = n$

$\frac{n!}{n_1! \dots n_r!} = \binom{n}{n_1, \dots, n_r}$ is called multinomial coefficient.

Connection to permutation with repetition:

Let $G = \{\underbrace{1, \dots, 1}_k, \underbrace{0, \dots, 0}_{n-k}\}$ be a multiset of n elements.
(Think about G as a set of name tags.)

An ordering of the elements of G

eg. 11010 ... 01
(1 2 ... j ... n)

$\longleftrightarrow$

A subset of k elements of H

$(h_1, h_2) / h_3, h_4 / h_5 \dots h_{n-1}, h_n$

$\longleftrightarrow$

(Choose the elements whose index is 1 in the permutation.)

$j=1 \Rightarrow h_j$ is chosen)

With repetition:

Let $H = \{\underbrace{h_{11}, \dots, h_{11}}_{n_1}, \underbrace{h_{21}, \dots, h_{21}}_{n_2}, \dots, \underbrace{h_{r1}, \dots, h_{r1}}_{n_r}\}$ be a multiset, and $k \leq n_i, \forall i \in \{1, 2, \dots, r\}$.

We can ~~define~~ determine each element of Ω by telling the number of h_1 's, h_2 's, ..., h_r 's it contains. Therefore

$$\Omega = \{(e_1, \dots, e_r) : e_i \geq 0, e_i \text{ is integer}, \sum_{i=1}^r e_i = k\}.$$

Prop.: $|\Omega| = \binom{k+r-1}{k}$

proof: We are going to use a combinatorial argument.

Suppose we have r urns named $e_1, e_2, \dots, e_r$, and we distribute k indistinguishable balls among them.

$\begin{array}{c} \bullet \bullet \\ e_1 \end{array} \begin{array}{c} | \\ \bullet \end{array} \begin{array}{c} | \\ \bullet \end{array} \dots \begin{array}{c} | \\ \bullet \end{array} \begin{array}{c} | \\ \bullet \end{array} \Rightarrow$ Each element of Ω is represented by ~~exactly~~ exactly one such distribution.

$\Rightarrow$ We just need to count the number of possible distributions!

Each distribution is a sequence of 2 characters:

$\begin{array}{c} \bullet \bullet \\ e_1 \end{array} \begin{array}{c} | \\ \bullet \bullet \bullet \end{array} \begin{array}{c} | \\ \bullet \end{array} \dots \begin{array}{c} | \\ \bullet \bullet \end{array} \begin{array}{c} | \\ \bullet \bullet \end{array} \leftarrow k \text{ (balls)} + r-1 \text{ (separators) characters in total}$

Thus the number of possible sequences is given by permutation with repetition: $\frac{(k+r-1)!}{(r-1)!k!} = \binom{k+r-1}{k}$

Or we can also count the total number of sequences using combination without repetition:

- we just choose the $r-1$ positions of the separators from the $k+r-1$ possibilities: $\binom{k+r-1}{k}$

11.1 Properties of binomial and multinomial coefficients

Thm.: (Pascal) Let $1 \leq n$ and $k \leq n$, then

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

proof: We are going to use a combinatorial argument.

Let $H = \{h_1, \dots, h_n\}$ be a set of n elements.

$\binom{n}{k}$ = number of k -element subsets of H
 $\uparrow$
 by definition

We can divide the subsets of H into 2 categories

$h_1 \in$ /

\ $h_1 \notin$

Number of k element subsets containing h_1 :

$\binom{n-1}{k-1} \leftarrow$ One element (h_1) is fixed. We choose the other $k-1$ from the remaining $n-1$ elements

Number of k element subsets that does not contain h_1 :

$\binom{n-1}{k} \leftarrow$ We choose all k elements from $\{h_2, \dots, h_n\}$ to avoid h_1 .

That is the number of k -element subsets of H is $\binom{n-1}{k-1} + \binom{n-1}{k}$. $\square$

Thm.: (Binomial Theorem) Let $x, y \in \mathbb{R}$ and $n \geq 1$, then

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

proof:

$$(x+y)^n = \underbrace{(x+y)(x+y) \cdots (x+y)}_{n \text{ times}}$$

After opening the brackets each element of the sum will have the form $x^k y^{n-k}$ for some $0 \leq k \leq n$.

What is the coefficient of $x^k y^{n-k}$ in the sum?

the n brackets $\rightarrow$ $\begin{matrix} \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \end{matrix}$
 $\begin{matrix} \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \end{matrix}$
 $\begin{matrix} \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } & \text{ } & \text{ } \end{matrix}$

We need to choose the k brackets where we chose x .

That is $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$.

$$\Rightarrow \binom{n}{k}$$

$\square$

Thm.: (Multinomial Theorem) Let $x_1, \dots, x_r \in \mathbb{R}$, and $n \geq 1$, then

$$(x_1 + \dots + x_r)^n = \sum_{\substack{\eta_1, \dots, \eta_r \geq 0 \\ \eta_1 + \dots + \eta_r = n}} \binom{n}{\eta_1, \dots, \eta_r} x_1^{\eta_1} \cdots x_r^{\eta_r}.$$

III. Basics of probability

We already defined the sample space (Ω) as the set of possible outcomes.

def. If Ω is finite or countable, then the subsets of Ω are called events.

If Ω is uncountable, then $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is called a set of possible events (or σ -algebra) if $\Omega \in \mathcal{F}$ $\leftarrow$ the set of all subsets (power set)

$$I) E \in \mathcal{F} \Rightarrow E^c := \Omega \setminus E \in \mathcal{F}$$

$$II) E_1, E_2, \dots \in \mathcal{F} \Rightarrow \bigcup_i E_i \in \mathcal{F}$$

Examples: 1) Race of 7 horses:

$$\Omega = \{\text{all permutations of 7 horses}\}$$

event: 5th horse wins the race

2) Tossing a coin: $\Omega = \{H, T\}$

Tossing two coins:

$$\Omega = \{(H, H), (H, T), (T, H), (T, T)\}$$

event: we get 'heads' at least once $\rightarrow E = \{(H, H), (H, T), (T, H)\}$

3) Tossing a dice until we roll a 6:

$$\Omega = \{(k_1, k_2, \dots, k_r, 6) : r \geq 0, k_i \in \{1, \dots, 5\} \text{ for all } i \leq r\} =$$

$$= \{(6), (1, 6), (2, 6), \dots, (5, 6), (1, 1, 6), (1, 2, 6), \dots\}$$

event: The result of the first toss is 4, and the third toss is 6.

$$E = \{(4, 1, 6), (4, 2, 6), (4, 3, 6), (4, 4, 6), (4, 5, 6)\}$$

def. $E, F \in \mathcal{F} \Rightarrow E \cup F = \{E \text{ or } F \text{ happened}\} = \{\omega \in \Omega : \omega \in E \text{ or } \omega \in F\}$

$E \cap F = \{E \text{ and } F \text{ happened}\} = \{\omega \in \Omega : \omega \in E \text{ and } \omega \in F\}$

def. The countably many events $E_1, E_2, \dots \in \mathcal{F}$ are called mutually exclusive if for all $i \neq j$: $E_i \cap E_j = \emptyset$.

def. We call a countable collection of events $E_1, E_2, \dots \in \mathcal{F}$ a partition or complete event system if they are mutually exclusive, and $\bigcup_i E_i = \Omega$.

Properties: - commutativity: $E \cup F = F \cup E$ and $E \cap F = F \cap E$
- associativity: $E \cup (F \cup G) = (E \cup F) \cup G$
 $E \cap (F \cap G) = (E \cap F) \cap G$
- distributivity: $E \cup (F \cap G) = (E \cup F) \cap (E \cup G)$
 $E \cap (F \cup G) = (E \cap F) \cup (E \cap G)$

Prop.: (De Morgan's rule)

$$a) \left(\bigcup_i E_i \right)^c = \bigcap_i E_i^c$$

$$b) \left(\bigcap_i E_i \right)^c = \bigcup_i E_i^c$$

proof: a)

$$\omega \in \left(\bigcup_i E_i \right)^c \Leftrightarrow \omega \notin \bigcup_i E_i \Leftrightarrow \forall i: \omega \notin E_i \Leftrightarrow \forall i: \omega \in E_i^c \Leftrightarrow \omega \in \bigcap_i E_i^c$$

b)

$$\omega \in \left(\bigcap_i E_i \right)^c \Leftrightarrow \omega \notin \bigcap_i E_i \Leftrightarrow \exists i: \omega \notin E_i \Leftrightarrow \exists i: \omega \in E_i^c \Leftrightarrow \omega \in \bigcup_i E_i^c \quad \square$$